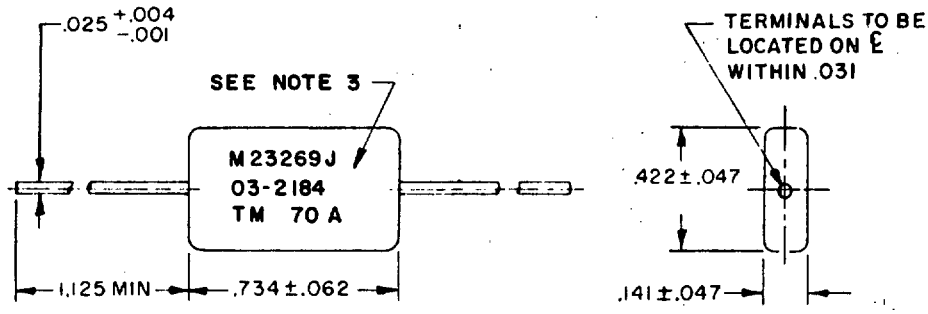
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, GLASS DIELECTRIC, (AXIAL WIRE-LEAD TERMINALS), ESTABLISHED RELIABILITY, STYLE CYR20

(E) FAILURE RATE LEVEL 'L' IS INACTIVE FOR DESIGN AFTER 20 July 1983.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-23269.



INCHES	MM
.001	.03
.004	.10
.025	.64
.031	.79
.047	1.19
.062	1.57
.141	3.58
.422	10.72
.734	18.64
1.125	28.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. See marking requirement.

FIGURE 1. Dimensions and configurations.

(E) denotes change.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:13Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case material: Glass.

Terminals: Axial wire-lead; solder coated, copper-clad, nickel-iron core (46 percent nickel, 54 percent iron).

Capacitance (nominal): See table I.

Capacitance tolerance: See table I.

Rated voltage: See table I.

Failure rate level: L, M, P, R, or S.

Rated temperature: -55°C to +125°C.

Temperature coefficient: 140 ±25 parts/million/°C.

Capacitance drift: 0.1 percent or 0.1 pF, whichever is greater.

Seal: Hermetic.

Seal: Test I.

Insulation resistance: 10,000 megohms or 200 ohm-farads minimum (qualification inspection only at +125°C). 100,000 megohms minimum (at +25°C).

Dissipation factor (DF): 0.1 percent maximum.

Thermal shock and moisture resistance:

DF: 0.1 percent maximum.

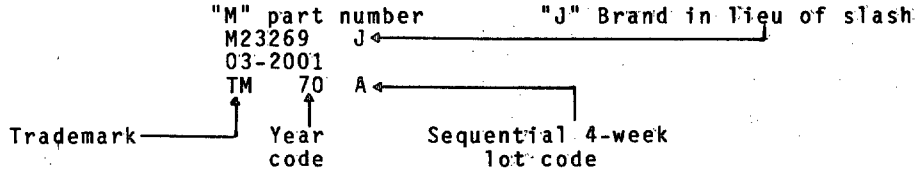
Quality factor (Q): In accordance with MIL-C-23269.

Life:

ΔCapacitance (240 hour and 2,000 hour): 0.5 percent maximum.

DF (240 hour and 2,000 hour): 0.15 percent maximum.

Marking:



Substitutability data: See table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:13Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical and physical characteristics and dash numbers.

Capacitance	Capacitance tolerance	Dash number M23269/03- FR level in percent/1,000 hours					
		500 V dc					100 V dc
		L (2.0)	M (1.0)	P (0.1)	R(0.01)	S (0.001)	S (0.001) 1/
(pF)							
560	±1%	2001	3001	4001	5001	6001	7001
560	±2%	2002	3002	4002	5002	6002	7002
560	±5%	2003	3003	4003	5003	6003	7003
620	±1%	2004	3004	4004	5004	6004	7004
620	±2%	2005	3005	4005	5005	6005	7005
620	±5%	2006	3006	4006	5006	6006	7006
680	±1%	2007	3007	4007	5007	6007	7007
680	±2%	2008	3008	4008	5008	6008	7008
680	±5%	2009	3009	4009	5009	6009	7009
750	±1%	2010	3010	4010	5010	6010	7010
750	±2%	2011	3011	4011	5011	6011	7011
750	±5%	2012	3012	4012	5012	6012	7012
820	±1%	2013	3013	4013	5013	6013	7013
820	±2%	2014	3014	4014	5014	6014	7014
820	±5%	2015	3015	4015	5015	6015	7015
910	±1%	2016	3016	4016	5016	6016	7016
910	±2%	2017	3017	4017	5017	6017	7017
910	±5%	2018	3018	4018	5018	6018	7018
1,000	±1%	2019	3019	4019	5019	6019	7019
1,000	±2%	2020	3020	4020	5020	6020	7020
1,000	±5%	2021	3021	4021	5021	6021	7021
1,100	±1%	2022	3022	4022	5022	6022	7022
1,100	±2%	2023	3023	4023	5023	6023	7023
1,100	±5%	2024	3024	4024	5024	6024	7024
1,200	±1%	2025	3025	4025	5025	6025	7025
1,200	±2%	2026	3026	4026	5026	6026	7026
1,200	±5%	2027	3027	4027	5027	6027	7027
1,300	±1%	2028	3028	4028	5028	6028	7028
1,300	±2%	2029	3029	4029	5029	6029	7029
1,300	±5%	2030	3030	4030	5030	6030	7030
1,500	±1%	2031	3031	4031	5031	6031	7031
1,500	±2%	2032	3032	4032	5032	6032	7032
1,500	±5%	2033	3033	4033	5033	6033	7033
1,600	±1%	2034	3034	4034	5034	6034	7034
1,600	±2%	2035	3035	4035	5035	6035	7035
1,600	±5%	2036	3036	4036	5036	6036	7036
1,800	±1%	2037	3037	4037	5037	6037	7037
1,800	±2%	2038	3038	4038	5038	6038	7038
1,800	±5%	2039	3039	4039	5039	6039	7039
2,000	±1%	2040	3040	4040	5040	6040	7040
2,000	±2%	2041	3041	4041	5041	6041	7041
2,000	±5%	2042	3042	4042	5042	6042	7042
2,200	±1%	2043	3043	4043	5043	6043	7043
2,200	±2%	2044	3044	4044	5044	6044	7044
2,200	±5%	2045	3045	4045	5045	6045	7045
2,400	±1%	2046	3046	4046	5046	6046	7046
2,400	±2%	2047	3047	4047	5047	6047	7047
2,400	±5%	2048	3048	4048	5048	6048	7048
2,700	±1%	2049	3049	4049	5049	6049	7049
2,700	±2%	2050	3050	4050	5050	6050	7050
2,700	±5%	2051	3051	4051	5051	6051	7051
3,000	±1%	2052	3052	4052	5052	6052	7052
3,000	±2%	2053	3053	4053	5053	6053	7053
3,000	±5%	2054	3054	4054	5054	6054	7054
3,300	±1%	2055	3055	4055	5055	6055	7055
3,300	±2%	2056	3056	4056	5056	6056	7056
3,300	±5%	2057	3057	4057	5057	6057	7057

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:13Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical and physical characteristics and dash numbers - Continued.

Capacitance	Capacitance tolerance	Dash number M23269/03- FR level in percent/1,000 hours					
		500 V dc					100 V dc
		L (2.0)	M (1.0)	P (0.1)	R(0.01)	S (0.001)	S (0.001) 1/
(pF)		300 V dc					100 V dc
3,600	±1%	2058	3058	4058	5058	6058	7058
3,600	±2%	2059	3059	4059	5059	6059	7059
3,600	±5%	2060	3060	4060	5060	6060	7060
3,900	±1%	2061	3061	4061	5061	6061	7061
3,900	±2%	2062	3062	4062	5062	6062	7062
3,900	±5%	2063	3063	4063	5063	6063	7063
4,300	±1%	2064	3064	4064	5064	6064	7064
4,300	±2%	2065	3065	4065	5065	6065	7065
4,300	±5%	2066	3066	4066	5066	6066	7066
4,700	±1%	2067	3067	4067	5067	6067	7067
4,700	±2%	2068	3068	4068	5068	6068	7068
4,700	±5%	2069	3069	4069	5069	6069	7069
5,100	±1%	2070	3070	4070	5070	6070	7070
5,100	±2%	2071	3071	4071	5071	6071	7071
5,100	±5%	2072	3072	4072	5072	6072	7072

1/ Sixth power rule applies for derating at 0.5 $\frac{\text{operating}}{\text{rated}}$ voltage ratios and below.

TABLE II. Substitutability data.

Superseded type designation in MIL-C-23269/3B	Use dash number M23269/03- FR level in percent/1,000 hours		Superseded type designation in MIL-C-23269/3B	Use dash number M23269/03- FR level in percent/1,000 hours	
	L (2.0)	M (1.0)		L (2.0)	M (1.0)
CYR20C561F	2001	3001	CYR20C182F	2037	3037
CYR20C561G	2002	3002	CYR20C182G	2038	3038
CYR20C561J	2003	3003	CYR20C182J	2039	3039
CYR20C621F	2004	3004	CYR20C202F	2040	3040
CYR20C621G	2005	3005	CYR20C202G	2041	3041
CYR20C621J	2006	3006	CYR20C202J	2042	3042
CYR20C681F	2007	3007	CYR20C222F	2043	3043
CYR20C681G	2008	3008	CYR20C222G	2044	3044
CYR20C681J	2009	3009	CYR20C222J	2045	3045
CYR20C751F	2010	3010	CYR20C242F	2046	3046
CYR20C751G	2011	3011	CYR20C242G	2047	3047
CYR20C751J	2012	3012	CYR20C242J	2048	3048
CYR20C821F	2013	3013	CYR20C272F	2049	3049
CYR20C821G	2014	3014	CYR20C272G	2050	3050
CYR20C821J	2015	3015	CYR20C272J	2051	3051
CYR20C911F	2016	3016	CYR20C302F	2052	3052
CYR20C911G	2017	3017	CYR20C302G	2053	3053
CYR20C911J	2018	3018	CYR20C302J	2054	3054
CYR20C102F	2019	3019	CYR20C332F	2055	3055
CYR20C102G	2020	3020	CYR20C332G	2056	3056
CYR20C102J	2021	3021	CYR20C332J	2057	3057
CYR20C112F	2022	3022	CYR20C362F	2058	3058
CYR20C112G	2023	3023	CYR20C362G	2059	3059
CYR20C112J	2024	3024	CYR20C362J	2060	3060
CYR20C122F	2025	3025	CYR20C392F	2061	3061
CYR20C122G	2026	3026	CYR20C392G	2062	3062

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:13Z
Check the source to verify that this is the current version before use.

TABLE II. Substitutability data - Continued.

Superseded type designation in MIL-C-23269/3B	Use dash number M23269/03- FR level in percent/1,000 hours		Superseded type designation in MIL-C-23269/3B	Use dash number M23269/03- FR level in percent/1,000 hours	
	L (2.0)	M (1.0)		L (2.0)	M (1.0)
CYR20C122J	2027	3027	CYR20C392J	2063	3063
CYR20C132F	2028	3028	CYR20C432F	2064	3064
CYR20C132G	2029	3029	CYR20C432G	2065	3065
CYR20C132J	2030	3030	CYR20C432J	2066	3066
CYR20C152F	2031	3031	CYR20C472F	2067	3067
CYR20C152G	2032	3032	CYR20C472G	2068	3068
CYR20C152J	2033	3033	CYR20C472J	2069	3069
CYR20C162F	2034	3034	CYR20C512F	2070	3070
CYR20C162G	2035	3035	CYR20C512G	2071	3071
CYR20C162J	2036	3036	CYR20C512J	2072	3072

Custodians:
Army - ER
Navy - EC
Air Force - 11
NASA - NA

Review activities:
Air Force - 17, 85
DLA - ES
ARMY - AV

User activities:
Navy - MC, CG, AS, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5910-1454-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:13Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:13Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command
Defense Standardization Program Branch
Department OF THE Navy
Washington, DC 20363

Attn: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:13Z
Check the source to verify that this is the current version before use.